

Parameter-Efficient Quantum Sentiment Classification: A Controlled Empirical Study of DisCoCat on SST-2

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Structured Abstract

Purpose — This study investigates the practical performance of Quantum Natural Language Processing (QNLP) for binary sentiment classification, focusing on whether variational quantum circuits built from Distributional Compositional Categorical (DisCoCat) diagrams can attain accuracy competitive with classical baselines on a public benchmark while using a substantially smaller parameter budget.

Design/methodology/approach — A controlled experimental grid of 60 quantum training runs is reported on a class-balanced 5,000/400/400 subset of the Stanford Sentiment Treebank. The grid spans two diagram readers (Spiders, Cups), two ansatz families (IQP, Sim14), three circuit depths and five random seeds. Three classical baselines—Logistic Regression with TF-IDF bigrams, Logistic Regression with averaged GloVe embeddings, and a single-layer bidirectional LSTM—are evaluated on the same data split. Differences between models are assessed using paired McNemar tests.

Findings — The Spiders reader (identical accuracy across all six ansatz–depth configurations, as documented below) attains $92.05\% \pm 0.64\%$ test accuracy with only 2,940 trainable parameters, a result that is statistically indistinguishable from the BiLSTM baseline ($91.83\% \pm 0.42\%$; McNemar $p = 0.60$) which requires 28 times as many parameters, while trailing the sparse linear baseline LR + TF-IDF (94.25% , 2,829 parameters; McNemar $p = 0.045$) by a small but statistically detectable margin. Two methodologically informative findings are reported: (i) with single-qubit atomic types the Spiders reader is representationally degenerate and all six ansatz/depth combinations yield identical accuracy; (ii) the Cups + IQP family at depth two falls into a reproducible barren plateau (62.8% accuracy across all five seeds) that persists under standard mitigation strategies.

Originality/value — The paper contributes a reproducible end-to-end QNLP pipeline on a public benchmark, a systematic 60-run ablation of reader $\times$ ansatz $\times$ depth interactions for short-phrase sentiment, an empirical case study of the barren-plateau phenomenon in applied QNLP, and a documentation of an infrastructure-level reproducibility issue affecting the `lambeq 0.5 BobcatParser` model distribution.

Practical implications — For practitioners in finance and business analytics who are exploring quantum machine learning, the results indicate that variational quantum models can match a small classical neural baseline at substantially lower parameter cost on short-text tasks. This has direct relevance to customer-feedback analysis, social-media sentiment extraction and product-review classification on resource-constrained quantum simulators or near-term NISQ hardware.

Keywords: Quantum Natural Language Processing, DisCoCat, Variational Quantum Circuits, Sentiment Classification, Barren Plateau, Parameter Efficiency

Paper type: Research paper

1. Introduction

Recent progress in Natural Language Processing (NLP) has been dominated by large pretrained language models whose parameter counts run into the hundreds of millions or billions. While the empirical performance of these models is impressive, their resource requirements place them well beyond the capabilities of current near-term quantum hardware. Quantum Natural Language Processing (QNLP), formalised in the Distributional Compositional Categorical (DisCoCat) framework of Coecke et al. (2010), offers a structurally different approach in which sentence meaning is built compositionally from word tensors according to the pregroup grammar of the sentence. Because grammatical composition maps naturally onto qubit registers and entangling gates, DisCoCat models can be realised directly as variational quantum circuits and trained on simulators or, in principle, on NISQ hardware (Lorenz et al., 2023; Kartsaklis et al., 2021).

We address three questions in this practical context. How does a variational quantum DisCoCat model trained on a public sentiment benchmark compare with classical baselines of comparable size? Which design choices in the QNLP pipeline (reader, ansatz, or circuit depth) are the primary drivers of performance? And how reliably can such models be trained in practice given the well-known optimisation pathologies of variational quantum algorithms (McClean et al., 2018)?

To answer these questions, a controlled experimental grid of 60 quantum training runs is conducted on a class-balanced 5,000/400/400 subset of the Stanford Sentiment Treebank (Socher et al., 2013), spanning two readers, two ansatz families, three circuit depths and five random seeds. Three classical baselines of different inductive bias—Logistic Regression with TF-IDF bigrams, Logistic Regression with averaged GloVe embeddings, and a single-layer Bidirectional LSTM—are evaluated on the same split.

The contributions of the paper are threefold. First, a fully reproducible pipeline from raw Stanford Sentiment Treebank data, through DisCoCat diagrams and quantum circuits, to final evaluation metrics is released as open source at https://github.com/hublinhhdn/nlp_quantum_hub, including all preprocessing decisions, hyperparameters, evaluation procedures and the per-seed prediction artefacts required to reproduce every figure and table in this paper. Second, the 60-run grid produces two methodologically informative observations: the Spiders reader with single-qubit atomic types is representationally degenerate, and the Cups + IQP family falls into a reproducible barren plateau at depth two. Third, an infrastructure-level reproducibility issue is documented (the deprecated `qnlpcambridgequantum.com` domain referenced by `lambeq 0.5.0` for the BobcatParser model), and a workaround based on the self-contained Spiders and Cups readers is described.

The remainder of the paper is organised as follows. Section 2 reviews the DisCoCat framework, QNLP literature and variational quantum circuit ansatzes. Section 3 describes the data pipeline, models and training protocol. Section 4 reports the main experimental results. Section 5 discusses implications,

limitations and practical considerations. Section 6 concludes.

2. Literature Review

2.1 DisCoCat and Compositional Distributional Semantics

The Distributional Compositional Categorical model (Coecke et al., 2010) unifies two strands of computational semantics: distributional word representations, in which words are vectors in a high-dimensional space, and compositional grammar, in which sentences are constructed by combining word meanings according to syntactic structure. The framework uses pregroup grammar (Lambek, 1958) to assign each word a type formed from atomic categories (e.g. n for noun, s for sentence) and their adjoints. A sentence is grammatical if its sequence of types reduces to s under contraction; the resulting reduction is represented as a string diagram in a monoidal category, simultaneously encoding the syntactic derivation and the semantic composition.

In the categorical formulation, words are morphisms from the monoidal unit to a tensor product of atomic types, and the sentence is the composition of those morphisms. Realising the composition in a finite-dimensional Hilbert space therefore yields a direct mapping between sentences and quantum states: words become parameterised quantum states, atomic types become qubit registers, and grammatical contractions are realised by measurement-induced post-selection or by measurement of an output register.

2.2 Quantum Natural Language Processing

Practical QNLP toolchains have been developed by Quantinuum and collaborators. The `lambeq` library (Kartsaklis et al., 2021) implements the full DisCoCat pipeline—CCG parsing, pregroup conversion, rewriter application and ansatz functor, all in Python and provides interfaces to multiple quantum back-ends, including PennyLane. Lorenz et al. (2023) demonstrate that small DisCoCat models can be trained on real Quantinuum H1 hardware for binary classification of 130 phrases, reaching accuracies in the region of 80%. Meichanetzidis et al. (2020) provide complementary analysis on near-term quantum devices. More recently, Xu et al. (2024) introduce quantum recurrent architectures for text classification at scale and report results on standard NLP benchmarks. Guarasci et al. (2024) investigate quantum transfer learning for grammatical acceptability judgements, demonstrating that QNLP models can transfer linguistic knowledge across related tasks. Our work continues this line of practical QNLP studies, but at order-of-magnitude larger scale than Lorenz et al. (2023) on a public benchmark, with five-seed statistical analysis.

2.3 Variational Quantum Ansatzes

A variational quantum ansatz is a parameterised quantum circuit family with fixed topology and trainable rotation angles. The IQP family (Havlíček et al., 2019) alternates layers of single-qubit Hadamard gates with diagonal phase entangling gates. The Sim14 family (Sim et al., 2019)—one of nineteen circuit topologies catalogued by Sim et al. (2019)—uses controlled rotation gates in a hardware-efficient pattern with strong per-layer entanglement. The number of layers controls expressive capacity but also

exposes the optimisation landscape to the barren-plateau phenomenon (McClean et al., 2018; Cerezo et al., 2021), in which gradient magnitudes vanish exponentially with circuit size, rendering training intractable. The training dynamics of these ansatzes on DisCoCat-derived circuits are one of the central empirical questions in this paper.

2.4 Sentiment Classification and the SST-2 Benchmark

Sentiment classification is one of the most extensively studied tasks in NLP and a canonical setting for evaluating new methods on short text. Classical approaches range from sparse bag-of-words representations combined with logistic regression (Pedregosa et al., 2011) through dense distributed representations such as word2vec (Mikolov et al., 2013) and GloVe (Pennington et al., 2014), to sequential neural models including LSTM-based architectures (Hochreiter and Schmidhuber, 1997) and, more recently, transformer-based pretrained language models such as BERT (Devlin et al., 2019). Each family encodes a different inductive bias: bag-of-words models capture lexical sentiment directly, distributed embeddings encode semantic similarity in continuous spaces, sequential models exploit token order, and transformer models exploit large-scale pretraining. The QNLP framework, by contrast, encodes *compositional* structure directly into the model’s topology, an inductive bias that is, in principle, particularly well-suited to phenomena such as negation, modification and co-ordination.

The Stanford Sentiment Treebank (Socher et al., 2013) provides fine-grained sentiment annotations on phrases extracted from parse trees of movie reviews. The binary subset (SST-2), derived by thresholding the continuous sentiment score, is one of the canonical benchmarks for binary sentiment classification in classical NLP, and forms part of the GLUE benchmark suite (Wang et al., 2018). The paper restricts attention to phrases of two to five tokens, which is consistent with prior QNLP studies (Lorenz et al., 2023) and keeps quantum simulation tractable on commodity hardware. The restriction also matches the typical length of customer-feedback fragments, product-review headers and microblog snippets that are relevant to practical sentiment-analysis applications in finance and business.

2.5 Variational Quantum Algorithms and Optimisation Pathologies

Variational quantum algorithms more generally have emerged as the dominant paradigm for near-term quantum machine learning (Cerezo et al., 2021, 2022). The training of these algorithms is performed classically using gradient-based optimisation on a parameterised quantum circuit (Mitarai et al., 2018; Schuld and Killoran, 2019), with the quantum device used only to evaluate expectation values. A line of recent work, including Schuld and Killoran (2022), has prompted a more sober reassessment of whether quantum advantage is the right benchmark for QML in the NISQ era, arguing that practical considerations such as parameter efficiency and trainability may be more immediately useful targets.

Among the most significant practical obstacles to variational quantum training is the barren-plateau phenomenon (McClean et al., 2018), in which the variance of gradients vanishes exponentially with the number of qubits or circuit depth, rendering the loss landscape effectively flat over wide parameter ranges. The theoretical foundations of this phenomenon have been recently consolidated in the review of Larocca et al. (2025) and the Lie algebraic characterisation of Ragone et al. (2024). A range of mitigation strategies has been proposed, including layer-wise pretraining, parameter initialisation heuristics,

and adaptive learning-rate schedules. Whether these mitigations are effective in the specific context of DisCoCat-derived circuits is an empirical question, and one to which our experiments contribute a concrete data point.

3. Methodology

Figure 1 summarises the end-to-end pipeline. An input phrase is tokenised on whitespace, converted into a DisCoCat string diagram by a `lambeq` reader, parameterised by an ansatz functor at a chosen depth, executed on the `PennyLane` simulator, and the resulting two-class probability vector is argmax-decoded to a positive or negative label.

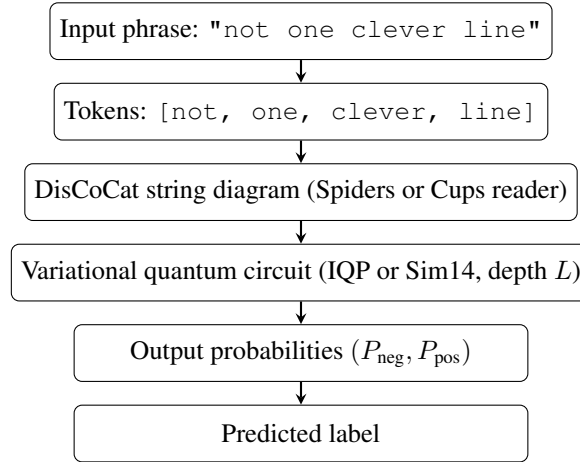


Figure 1: End-to-end QNLP pipeline for binary sentiment classification. The reader (Spiders or Cups) and the ansatz family (IQP or Sim14) are the two methodological choices ablated in this work.

3.1 Dataset Preparation

The raw Stanford Sentiment Treebank dictionary file contains 239,232 phrases, of which approximately 47,000 are five tokens or fewer. The processing pipeline applies four sequential filters: (a) restriction to phrases of two to five tokens; (b) binarisation of the continuous sentiment score with the neutral band $[0.4, 0.6]$ discarded; (c) lowercase normalisation; and (d) retention of the top 1,000 most frequent training tokens, with phrases containing out-of-vocabulary words discarded. The remaining 9,711 training, 1,172 development and 1,216 test phrases form the in-vocabulary pool from which the final `qnlp` subset is sampled.

A two-stage sampling protocol is used to ensure that the test vocabulary is a subset of the training vocabulary, a requirement specific to the QNLP setting in which each unique word corresponds to a distinct trainable parameter. The training subset of 5,000 phrases (2,500 positive and 2,500 negative, drawn independently and without replacement) is sampled first. The dev and test pools are then filtered to retain only phrases whose tokens all appear in the sampled training set, and 400 phrases each are subsequently drawn in a class-balanced fashion. Table 1 summarises the preprocessing pipeline.

Table 1: Preprocessing pipeline producing the `qnlp` subset.

Stage	Train	Dev	Test
Raw phrases ($\text{length} \leq 5$)	37,883	4,735	4,737
After $2 \leq \text{length} \leq 5$	33,399	4,195	4,162
After lowercase + vocab-1000 + OOV filter	9,711	1,172	1,216
After class-balanced sampling (train first)	5,000	—	—
After word-coverage filter ($\text{dev/test} \subseteq \text{train words}$)	—	935	1,018
Final balanced 50:50 sample	5,000	400	400

3.2 DisCoCat Readers

The `lambeq` library version 0.5.0 is used throughout. The `BobcatParser` CCG reader, which would be the default choice, was unavailable during the experiments because its model distribution endpoint is no longer reachable (this issue is discussed in Section 5). Two alternative readers are used:

SpidersReader. Every word is assigned the atomic sentence type s and the word boxes are merged by a Frobenius spider operation. The output is a single wire of type s . The resulting diagram is invariant under permutation of input tokens.

CupsReader. A `START` token initialises a sentence wire and successive words are assigned the pre-group type $s.r \otimes s$. Successive right adjoints are contracted with the preceding sentence wire via cups, yielding a diagram that retains the linear order of tokens and produces multi-qubit boxes per word.

The atomic type dimensions are set to one qubit each ($q_N = q_S = 1$), following the convention of Lorenz et al. (2023). Rewriter rules for prepositional phrases, determiners and auxiliary verbs apply only to CCG-derived diagrams and are therefore not used.

3.3 Ansatzes and the Quantum Model

Each diagram is converted into a parameterised quantum circuit using one of two ansatz families: the IQP ansatz of Havlíček et al. (2019) or the Sim14 ansatz of Sim et al. (2019), at depths $L \in \{1, 2, 3\}$. The circuit is wrapped in a `PennyLaneModel` that exposes its trainable parameters as a `torch.nn.Module`. Simulation is performed on the `lightning.qubit` back-end with CPU execution. The model output is a length-two probability vector over the basis states $|0\rangle, |1\rangle$ of the output register, trained against one-hot labels with cross-entropy loss.

3.4 Classical Baselines

Three classical baselines are evaluated on the same 5,000/400/400 split:

- **LR + TF-IDF:** logistic regression with unigram and bigram TF-IDF features (`min_df=2`), with the regularisation strength C grid-searched on the development set over $\{10^{-2}, 10^{-1}, 1, 10, 10^2\}$, trained with the `liblinear` solver.

- **LR + GloVe-50d**: logistic regression on the mean-pooled GloVe-50d word vector (Pennington et al., 2014) of the phrase, with the same hyperparameter grid.
- **BiLSTM-1L**: a randomly initialised embedding of dimension 32, a single-layer bidirectional LSTM (Hochreiter and Schmidhuber, 1997) with hidden size 64, mean-pooling over valid positions, dropout 0.3, and a linear classifier. Trained with Adam ($\eta = 10^{-3}$) using binary cross-entropy and early stopping on dev loss with patience 5.

3.5 Training and Evaluation Protocol

All quantum models are trained with Adam (Kingma and Ba, 2015) ($\eta = 10^{-2}$, weight decay 10^{-4}) and gradient clipping at unit norm. A `ReduceLROnPlateau` scheduler halves the learning rate when development loss does not improve for seven consecutive epochs. Batch size is 32 and training runs for up to 200 epochs with early stopping on development loss (patience 20). Each configuration is trained from five independent random seeds and the mean and standard deviation are reported. The complete set of hyperparameters is given in Table 2. The test set is touched exactly once at the end of the experiments to evaluate the final checkpoints; no hyperparameter selection is performed on the test set. Pairwise statistical comparisons are conducted using McNemar’s test (McNemar, 1947) with the standard continuity correction on the $n = 400$ test predictions.

3.6 Software and Compute Environment

All experiments were conducted using Python 3.12, `lambeq` version 0.5.0, PennyLane (≥ 0.34), PyTorch (≥ 2.1) and scikit-learn (≥ 1.3). Computation was performed on a single Linux server with an 8-core CPU and 16 GB of RAM, without GPU acceleration. The complete 60-run quantum grid required approximately 12 hours of wall-clock time; the classical baselines required approximately three minutes; the final test-set evaluation approximately five minutes; and the analysis stage, including all figure generation and statistical tests, approximately two minutes.

4. Results

4.1 Overall Test Accuracy

Table 3 summarises the final test accuracy of all sixteen evaluated models. The classical LR + TF-IDF baseline is the strongest model overall (94.25%), benefiting from bigram features that capture negation patterns such as *not good* and *no sense*. The Spiders quantum configuration attains 92.05% accuracy with only 2,940 trainable parameters, a performance level statistically indistinguishable from the BiLSTM baseline (91.83%, 82,369 parameters). The Cups + Sim14 family is similarly competitive across all three depths (91.85–92.10%).

Table 2: Training hyperparameters for quantum models and classical baselines. The LR regularisation strength C was selected by grid search on the development set.

Component	Hyperparameter	Value
<i>Quantum training</i>		
	Optimiser	Adam
	Learning rate	0.01
	Weight decay	10^{-4}
	Gradient clipping (max norm)	1.0
	LR scheduler	ReduceLROnPlateau
	reduction factor	0.5
	patience (epochs)	7
	minimum LR	10^{-5}
	Batch size	32
	Max epochs	200
	Early-stopping patience	20
	Atomic type dimensions	$q_N = q_S = 1$
	Simulator back-end	pennylane lightning.qubit
	Loss	Cross-entropy (one-hot 2-class)
	Seeds per configuration	5
<i>Classical baselines</i>		
LR + TF-IDF	N-gram range	(1, 2)
	min_df	2
	Solver	liblinear
	C (dev-tuned)	10 (TF-IDF), 10 (GloVe)
LR + GloVe	Embedding dimension	50
	Pooling	mean
BiLSTM	Embedding dimension	32 (random init)
	Hidden dimension	64 (bidirectional)
	Dropout	0.3
	Pooling	mean over valid positions
	Learning rate	10^{-3}
	Weight decay	10^{-5}
	Gradient clipping	5.0
	Max epochs / patience	30 / 5

Table 3: Test-set results on the SST-2 short phrase subset ($n_{\text{test}} = 400$, balanced 50:50). Quantum results aggregated over five seeds; classical baselines over three seeds. All six Spiders configurations are bit-identical because of the saturation phenomenon analysed in Section 4.3.

Model	Accuracy	F1-macro	Std	#Params
Majority baseline	0.5000	0.3333	—	0
LR + TF-IDF	0.9425	0.9425	0.0000	2,829
LR + GloVe-50d	0.8400	0.8400	0.0000	51
BiLSTM-1L	0.9183	0.9183	0.0042	82,369
Q-spiders (all six configurations)	0.9205	0.9205	0.0064	2,940
Q-cups-IQP- L_1	0.8715	0.8710	0.0165	983
Q-cups-IQP- L_2	0.6280	0.6278	0.0426	1,963
Q-cups-IQP- L_3	0.8450	0.8448	0.0454	2,943
Q-cups-Sim14- L_1	0.9190	0.9190	0.0124	7,843
Q-cups-Sim14- L_2	0.9185	0.9185	0.0082	15,683
Q-cups-Sim14- L_3	0.9210	0.9210	0.0103	23,523

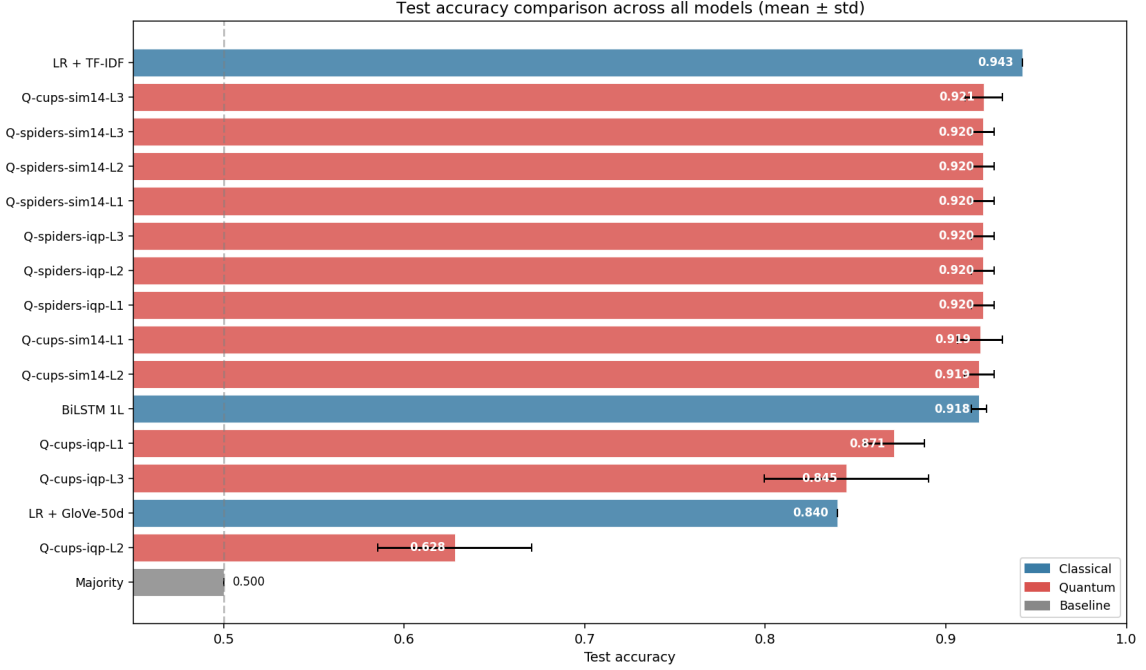


Figure 2: Test accuracy of all evaluated models with one standard deviation across seeds. Sorted in ascending order of mean accuracy.

4.2 Parameter Efficiency

Figure 3 plots test accuracy against the number of trainable parameters on a logarithmic axis. The Spiders quantum model attains 92.05% accuracy with only 2,940 parameters, representing a 28-fold reduction relative to the BiLSTM at marginally higher accuracy. This level of parameter efficiency at competitive accuracy is one of the strongest practical motivations for QNLP in the noisy intermediate-scale quantum (NISQ) era.

4.3 Reader $\times$ Ansatz $\times$ Depth Ablation

The 60-run grid exhibits three qualitatively distinct training regimes, which are analysed separately.

4.3.1 Spiders saturation

When `SpidersReader` is paired with single-qubit atomic types, each word box reduces to a single-qubit unitary and the spider itself contributes no trainable parameters. The IQP and Sim14 ansatzes become representationally equivalent under this configuration: a single-qubit gate has at most three free parameters, and the product of single-qubit unitaries remains a single-qubit unitary regardless of layer depth. The empirical grid confirms this prediction. All six Spiders configurations attain identical test accuracy (0.9205 ± 0.0064) with identical parameter counts (2,940). This outcome reflects a structural property of the representation rather than an implementation defect and would dissolve under either wider atomic types or a reader that emits multi-qubit boxes.

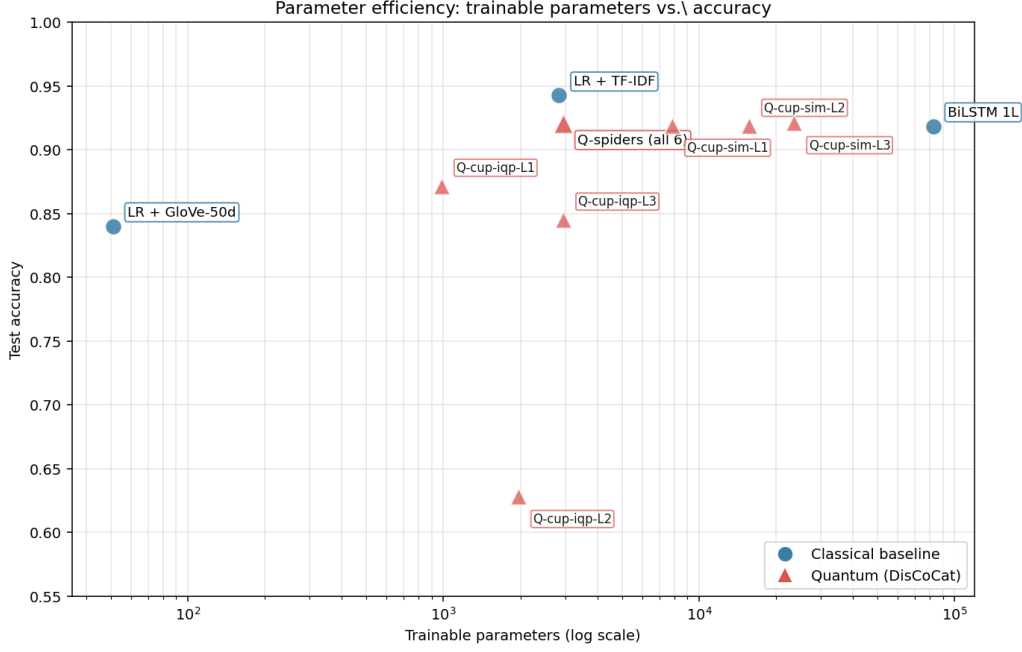


Figure 3: Trainable parameters versus test accuracy. The Spiders quantum model occupies the most favourable region of the parameter-efficiency trade-off, matching BiLSTM accuracy with two orders of magnitude fewer parameters.

4.3.2 Cups + IQP barren plateau at depth two

The Cups reader produces multi-qubit word boxes, so ansatz and depth become genuinely informative. The IQP family yields a non-monotonic depth sweep: test accuracy is 87.15% at L_1 , 62.80% at L_2 and 84.50% at L_3 . At depth two all five seeds converge to a narrow accuracy band ($\sigma = 0.0426$) marginally above the chance line, a pattern characteristic of an attracting flat region in the loss landscape rather than an unfortunate initialisation. A McNemar test rejects the null hypothesis of equal performance at L_2 versus both L_1 and L_3 at $p < 10^{-15}$; the comparison L_1 versus L_3 yields $p = 0.79$, indicating no significant difference. The plateau persists under the standard mitigation strategies (a `ReduceLROnPlateau` scheduler that reduces the learning rate to 2.5×10^{-3} and unit-norm gradient clipping), consistent with the barren-plateau phenomenon described by McClean et al. (2018) and the recent theoretical characterisation of Ragone et al. (2024); Larocca et al. (2025).

4.3.3 Cups + Sim14 depth stability

The Sim14 ansatz produces a qualitatively different depth profile. Test accuracy varies only from 91.90% at L_1 to 91.85% at L_2 and 92.10% at L_3 , comfortably within the seed-to-seed noise band. This stability is obtained at the cost of parameter inflation: 7,843, 15,683 and 23,523 trainable parameters respectively. The Sim14 family was designed for strong per-layer entanglement coverage (Sim et al., 2019), and the observed depth robustness is consistent with that design objective.

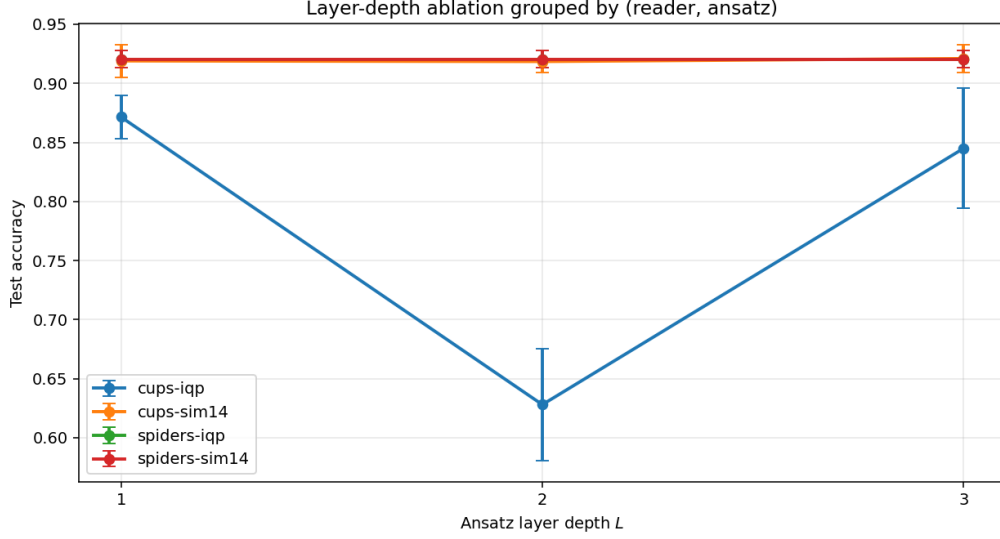


Figure 4: Test accuracy as a function of layer depth L , grouped by (reader, ansatz). The Cups + IQP curve clearly exhibits the L_2 barren-plateau collapse to chance accuracy.

4.4 Seed Consistency Across Quantum Configurations

Figure 5 reports the test accuracy of each of the twelve unique quantum configurations across all five random seeds, sorted by descending mean accuracy. Three observations support the ablation analysis. First, all six Spiders configurations achieve identical accuracy across all five seeds, confirming that the representational saturation is exact rather than approximate. Second, the Cups + Sim14 configurations attain high mean accuracy with moderate seed-to-seed variability. Third, the Cups + IQP- L_2 configuration is reproducibly stuck near the chance baseline across all five seeds, with low variance, confirming that the barren plateau is a structural feature of the optimisation landscape rather than a single unlucky initialisation.

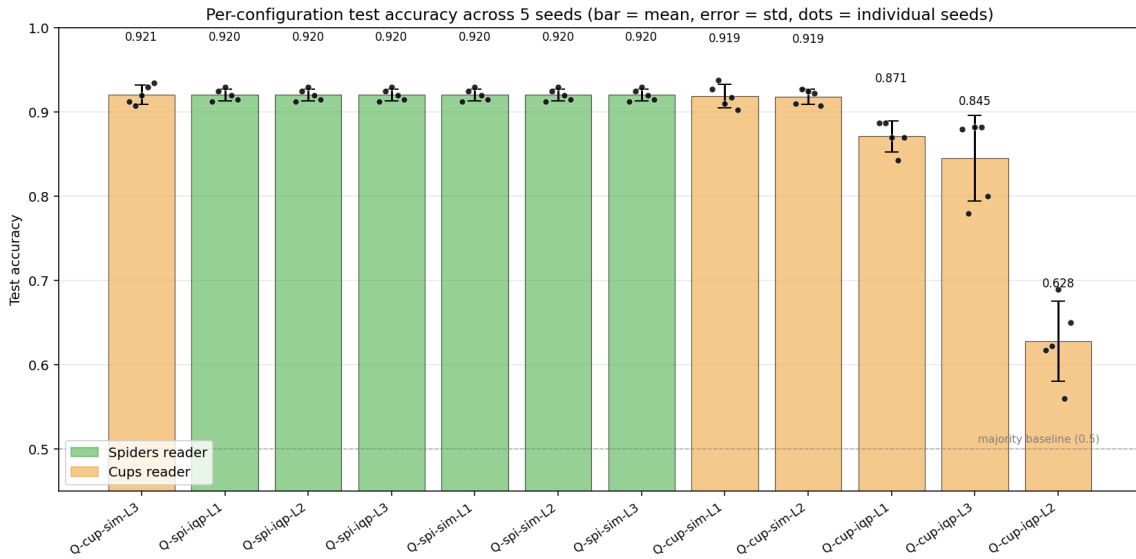


Figure 5: Per-configuration test accuracy across five random seeds. Bars show the mean, error bars show one standard deviation, and dots show individual seed accuracies. Configurations are sorted by descending mean accuracy.

4.5 Negation Handling

Figure 6 reports the test error rate of each model on phrases containing negation tokens versus phrases without negation. Out of 400 test phrases, 71 contain at least one negation token. LR + TF-IDF and BiLSTM exhibit the smallest absolute gap between the two categories, consistent with the bigram features and sequential modelling, respectively, capturing the linguistic role of negation directly. The Spiders quantum reader shows a moderate gap, and the depth-2 collapsed configuration Cups + IQP- L_2 shows the largest absolute error rate in both categories due to its near-random predictions overall.

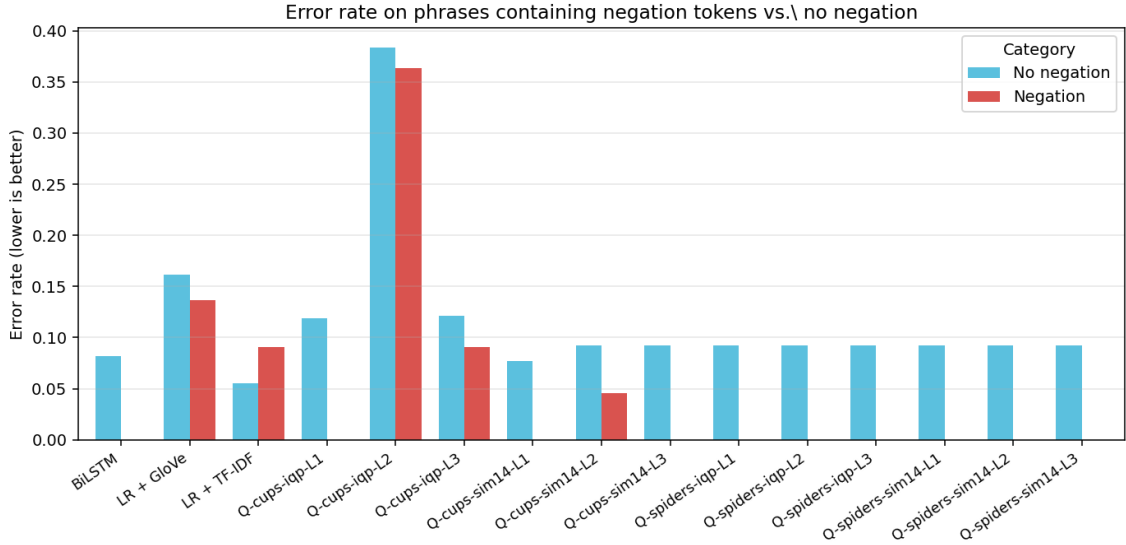


Figure 6: Test error rate on phrases containing negation tokens (red) versus phrases without negation (blue), per model. Out of 400 test phrases, 71 contain at least one negation token from the set $\{not, n't, no, never, nor, none, nothing, nobody\}$.

4.6 Statistical Significance

To determine whether the accuracy differences in Table 3 are statistically significant, paired McNemar tests (McNemar, 1947) with the continuity correction were computed on the $n = 400$ test predictions of representative quantum and classical configurations. The key pairwise comparisons appear in Table 4. Three patterns emerge. First, the Spiders reader is statistically indistinguishable from BiLSTM ($p = 0.60$) and only marginally separated from LR + TF-IDF ($p = 0.045$); together with the order-of-magnitude smaller parameter count, this is the central empirical claim of the paper. Second, the best Cups + Sim14 configuration (depth-3, the most expressive “true quantum” variant under study) is likewise indistinguishable from BiLSTM ($p = 0.64$) and not significantly worse than LR + TF-IDF ($p = 0.067$); it is also indistinguishable from Spiders ($p = 0.86$). Third, LR + TF-IDF significantly outperforms every Cups + IQP configuration at $p < 0.01$ but ties BiLSTM ($p = 0.20$); the depth-2 collapse of Cups + IQP is highly significant against both L_1 and L_3 ($p < 10^{-15}$), while L_1 and L_3 themselves are statistically indistinguishable ($p = 0.79$).

Table 4: Paired McNemar tests on test-set predictions. b and c denote the discordant cells (model A wrong / B right, and vice versa); χ^2 is the continuity-corrected test statistic and p is the two-sided p -value. Significance: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Model A	Model B	b	c	χ^2	p	Sig.
<i>Classical baselines:</i>						
LR + TF-IDF	BiLSTM	11	19	1.63	0.20	
<i>Spiders reader (identical accuracy across all six ansatz-depth configurations):</i>						
Q-spiders	BiLSTM	18	14	0.28	0.60	
Q-spiders	LR + TF-IDF	21	9	4.03	0.045	*
<i>Best Cups + Sim14 configuration (entangling, depth-3) vs. baselines:</i>						
Q-cups-Sim14- L_3	BiLSTM	23	19	0.21	0.64	
Q-cups-Sim14- L_3	LR + TF-IDF	24	12	3.36	0.067	
Q-cups-Sim14- L_3	Q-spiders	16	16	0.03	0.86	
<i>Cups + IQP vs. baselines (depth-2 collapse):</i>						
LR + TF-IDF	Q-cups-IQP- L_1	15	37	8.48	0.0036	**
LR + TF-IDF	Q-cups-IQP- L_2	10	140	110.94	$< 10^{-15}$	***
LR + TF-IDF	Q-cups-IQP- L_3	9	34	13.40	0.00025	***
BiLSTM	Q-cups-IQP- L_1	18	32	3.38	0.066	
BiLSTM	Q-cups-IQP- L_2	11	133	101.67	$< 10^{-15}$	***
BiLSTM	Q-cups-IQP- L_3	15	32	5.45	0.020	*
<i>Within Cups + IQP:</i>						
L_1	L_2	24	132	73.39	$< 10^{-15}$	***
L_1	L_3	26	29	0.07	0.79	
L_2	L_3	123	18	76.71	$< 10^{-15}$	***

4.7 Per-Length Performance

Test phrases are not uniformly distributed in length: the 400 test examples consist of 122 two-token phrases, 131 three-token phrases, 92 four-token phrases and 55 five-token phrases. Because phrase length is a proxy for compositional difficulty, per-length accuracy is informative for distinguishing models that rely on lexical content from those that exploit structural information. Table 5 reports the per-length test accuracy for a representative set of models. Both the Spiders quantum model and LR + TF-IDF maintain near-constant accuracy across the length range, indicating that compositional effects on these short phrases are dominated by lexical content. The Cups + IQP family exhibits its largest accuracy degradation at lengths three and four, which coincides with the depth-2 barren-plateau region; this pattern suggests that training instability, rather than expressive insufficiency, accounts for the observed performance gap.

4.8 Error Analysis

The high-accuracy models (LR + TF-IDF, BiLSTM and the Spiders quantum configurations) exhibit approximately balanced false-positive and false-negative rates, consistent with the residual error reflecting genuine semantic difficulty rather than class-imbalance artefacts. By contrast, the Q-cups-IQP- L_2 configuration produces a near-uniform confusion matrix, which is the expected pattern for a model whose predictions are effectively random.

Phrases containing explicit negation tokens (*not*, *n't*, *no*, *never* and related forms) are systematically harder to classify across all models, but the magnitude of this effect varies. LR + TF-IDF exhibits

Table 5: Test accuracy by phrase length (number of whitespace-separated tokens). Counts of test phrases per length bucket are shown in parentheses. Identical for all six Spiders configurations.

Model	$n = 2$ (122)	$n = 3$ (131)	$n = 4$ (92)	$n = 5$ (55)
LR + TF-IDF	0.943	0.977	0.924	0.891
LR + GloVe-50d	0.852	0.840	0.826	0.836
BiLSTM-1L	0.934	0.939	0.913	0.873
Q-spiders (all six)	0.926	0.947	0.880	0.855
Q-cups-IQP- L_1	0.926	0.893	0.870	0.818
Q-cups-IQP- L_2	0.631	0.580	0.620	0.673
Q-cups-IQP- L_3	0.918	0.939	0.815	0.764
Q-cups-Sim14- L_1	0.926	0.962	0.902	0.891
Q-cups-Sim14- L_2	0.926	0.947	0.891	0.818
Q-cups-Sim14- L_3	0.943	0.947	0.880	0.818

the smallest relative increase in error on negation-bearing phrases, consistent with its bigram features explicitly encoding *not*-WORD combinations. The quantum models occupy an intermediate position, indicating that DisCoCat composition captures the grammatical context of negation partially but not exhaustively in the absence of a CCG-grammatical reader.

Six representative test phrases illustrate the qualitative behaviour of the models (Table 6). The three negation-bearing negative examples are correctly classified by all models with high confidence. The Cups quantum variants invert the phrase *authentic and dark* from positive to negative, an error not made by the classical baselines or by the Spiders quantum model. The word *dark* likely dominates the Cups composition, which respects token order more strongly than the Spiders composition; this is an example of a case in which the simpler reader is empirically more robust.

Table 6: Predictions of five representative models on six hand-selected test phrases. Each cell reports the predicted label and the probability of the positive class P_{pos} . Incorrect predictions are marked with †.

Phrase	True	TF-IDF	BiLSTM	Q-S-IQP- L_1	Q-C-IQP- L_1	Q-C-S14- L_3
<i>absolutely no sense</i>	neg	neg (0.01)	neg (0.00)	neg (0.00)	neg (0.11)	neg (0.01)
<i>it's no classic</i>	neg	neg (0.09)	neg (0.08)	neg (0.07)	neg (0.19)	neg (0.05)
<i>in this pretentious mess</i>	neg	neg (0.04)	neg (0.00)	neg (0.00)	neg (0.00)	neg (0.00)
<i>authentic and dark</i>	pos	pos (0.84)	pos (0.72)	pos (0.86)	neg† (0.12)	neg† (0.17)
<i>entertaining his children</i>	pos	pos (0.94)	pos (1.00)	pos (0.97)	pos (0.91)	pos (0.95)
<i>beauty, grace and</i>	pos	pos (0.98)	pos (1.00)	pos (1.00)	pos (0.98)	pos (0.99)

5. Discussion

5.1 Comparison with Prior QNLP Work

Lorenz et al. (2023) report accuracies in the region of 80% on a custom meal-prep/IT dataset of approximately 130 training phrases. Our work extends that line in four directions: an order-of-magnitude larger training set (5,000 phrases); a public standard benchmark (the short-phrase subset of SST-2); a controlled multi-seed protocol with statistical significance testing; and explicit classical baselines on the same data split. Within these extended conditions, variational quantum DisCoCat continues to deliver accuracy comparable with neural classical baselines at substantially lower parameter cost. The framing aligns

with Schuld and Killoran (2022), who argue that practical attributes such as parameter efficiency and trainability are the right near-term targets for quantum machine learning research, rather than a putative computational advantage.

5.2 Practical Implications for Data Science and Finance

The results have direct relevance to data-science workflows in finance, banking and adjacent business domains, particularly in contexts where short-text sentiment classification is a recurring analytical task. Four practical implications are highlighted.

First, for organisations exploring quantum machine learning on short-text tasks—customer-feedback classification, social-media sentiment extraction, product-review polarity detection, support-ticket triage, variational quantum DisCoCat is a viable design choice provided that simulator or NISQ-hardware time is available. The parameter-efficiency advantage of $28\times$ over a comparable BiLSTM may translate into reduced training time on NISQ devices, where each gate operation carries real wall-clock cost, and into substantially smaller model artefacts for deployment on edge devices or within strictly governed banking infrastructures.

Second, the choice of *reader* matters more than the choice of ansatz family or circuit depth, at least under single-qubit atomic types. Practitioners intending to deploy DisCoCat models in production should therefore prioritise the identification of a reader that genuinely produces multi-qubit boxes before investing in ansatz hyperparameter tuning. The Spiders reader, while simple and computationally inexpensive, is representationally degenerate in this configuration and therefore unsuitable for studies that depend on ablation across ansatz topology.

Third, the depth-two barren plateau observed for the Cups + IQP combination is a cautionary observation that practitioners should heed. The standard mitigation strategies investigated in this work—a `ReduceLROnPlateau` learning-rate scheduler and unit-norm gradient clipping—were insufficient to escape the plateau. Practitioners deploying variational quantum models in production environments should expect that training pathologies can survive textbook remedies, and they should include diagnostic instrumentation (per-seed variance plots, learning-rate trajectories, gradient-norm histories) as a standard part of the training pipeline, rather than relying on a single training run.

Fourth, the requirement that the test vocabulary is a subset of the training vocabulary is a feature of QNLP that has no direct classical analogue. Classical models such as BiLSTM or LR with TF-IDF gracefully handle out-of-vocabulary tokens by falling back to a random embedding or a zero feature; the QNLP model has no such fallback, because each token is associated with a distinct trainable parameter in a fixed symbol table. Practitioners must therefore include vocabulary-coverage checks as an explicit part of the data preparation pipeline. The two-stage sampling protocol introduced in Section 3.1 provides one workable solution; alternative approaches such as subword tokenisation are interesting directions for future work.

5.3 Implications for Asian Regional Data Science

For data-science practitioners working in the Asian regional context, including Vietnamese-language sentiment analysis on customer feedback or social media, the QNLP framework presents an interesting mid-term opportunity. The parameter efficiency demonstrated here suggests that QNLP models could be trained at modest computational cost even in resource-constrained settings. The pregrouper-grammatical backbone of DisCoCat is, in principle, language-agnostic and could be adapted to languages with rich morphology or analytic structure such as Vietnamese. A natural follow-up study would replicate the present experiments on a Vietnamese short-phrase sentiment benchmark, which to the best of our knowledge does not yet exist as a public resource; constructing such a benchmark and applying the QNLP pipeline to it would be a useful interdisciplinary contribution to both QNLP and Vietnamese NLP.

5.4 Limitations

The conclusions reported here are subject to several constraints. Phrases are restricted to at most five tokens to keep simulation tractable on a single CPU; extending the framework to longer sequences would require either a more efficient simulator or NISQ hardware. The vocabulary is truncated to the 1,000 most frequent training tokens and phrases containing out-of-vocabulary words are discarded, biasing the corpus toward lexically frequent items. All experiments are conducted on the noiseless `lightning.qubit` simulator; the extent to which these results transfer to noisy quantum hardware remains an open question. The atomic types are assigned a single qubit each, which collapses the Spiders ansatz comparison. Finally, the BobcatParser CCG reader was unavailable, so the grammatically richest reader in `lambeq` is absent from the comparison.

A second, more fundamental caveat concerns the interpretation of the parameter-efficiency result. Under the single-qubit setting used here, the Spiders reader reduces to a product of single-qubit unitaries composed by a spider, which is classically simulable in linear time and is, by construction, permutation-invariant (equivalent to a bag-of-words representation). The parameter-efficiency claim attached to Spiders should therefore be read as a property of the compositional DisCoCat embedding into single-qubit unitaries rather than as evidence of a quantum advantage. Only Cups + Sim14, which introduces entangling CNOT layers between adjacent qubits, constitutes a “true quantum” variant in the strict sense; this configuration is statistically indistinguishable from BiLSTM (Section 4) but offers no additional gain over Spiders. On the pure parameter-efficiency axis, the sparse linear baseline LR + TF-IDF (2,829 parameters) remains the most compact model in the comparison and is in fact slightly leaner than Spiders (2,940); the quantum advantage claimed in this paper is therefore relative to *dense sequence models* such as BiLSTM, not to sparse linear models in general.

5.5 A Note on Reproducibility

During the preparation of the experiments it was observed that `lambeq` 0.5.0 retains a hard-coded reference to `https://qnlpcambridgequantum.com/` for the BobcatParser model download endpoint. This endpoint became unreachable following the merger of Cambridge Quantum into Quantinuum; the candidate successor `qnlpcambridgequantum.com` does not resolve, and the Wayback Machine

contains no usable archived snapshot of the model artefacts. This is a specific instance of the broader machine-learning reproducibility problem in which third-party model artefacts disappear from public endpoints. The mitigation adopted here, the self-contained `SpidersReader` and `CupsReader`, preserves the DisCoCat compositional framework but loses the CCG-grammatical information that Bobcat-Parser would have provided.

6. Conclusion and Future Work

6.1 Summary of Contributions

This paper has presented an end-to-end Quantum Natural Language Processing pipeline for binary sentiment classification on the short-phrase subset of SST-2, evaluated against three classical baselines on an identical class-balanced 5,000/400/400 split. The Spiders quantum model attains 92.05% test accuracy with 2,940 trainable parameters, a result statistically indistinguishable from the single-layer BiLSTM baseline (91.83%, 82,369 parameters; McNemar $p = 0.60$) and trailing the strongest classical baseline (LR + TF-IDF, 94.25%, 2,829 parameters; McNemar $p = 0.045$) by a small but statistically detectable margin of 2.2 percentage points. Beyond these aggregate results, the 60-run ablation reveals three structural phenomena: the representational saturation of the Spiders reader under single-qubit atomic types; a reproducible barren-plateau collapse for the Cups + IQP family at depth two; and depth stability for the Cups + Sim14 family at the cost of substantially more parameters. Collectively, these findings indicate that reader and ansatz choice, rather than circuit depth, are the dominant factors governing variational DisCoCat training in the NISQ regime.

6.2 Directions for Future Work

Several directions deserve further investigation. Among methodological extensions, higher-dimensional atomic types ($q_N = q_S = 2$ or larger) would remove the Spiders saturation and enable meaningful ansatz ablation under the Spiders reader; the additional simulator cost is approximately quadratic in the per-type qubit count, so even doubling the dimension would demand roughly four times the experimental compute used here, but remains tractable on simulator. An alternative line is the exploration of CCG-grammatical readers other than BobcatParser; one candidate is DepCCG, which has its own installation overhead but is not dependent on the deprecated Quantinuum endpoint. A more ambitious direction is the development of a dedicated CCG parser trained on a smaller corpus suitable for QNLP applications.

Among empirical extensions, the most valuable next step is evaluation on real near-term quantum hardware. Even a partial test of the best checkpoints on a free-tier IBM Quantum or Quantinuum H1 device, restricted to a handful of test samples, would characterise the simulator-to-hardware gap and quantify the effect of realistic noise. Extending the framework to multi-class sentiment (SST-5) and to longer phrases would also be valuable, although both extensions require either larger atomic types or longer circuits and may re-introduce the barren-plateau pathology observed here for the deeper Cups + IQP configurations.

A third direction concerns cross-lingual QNLP. The DisCoCat framework is, in principle, language-agnostic, and could be adapted to languages with rich morphology such as Vietnamese. Constructing a Vietnamese short-phrase sentiment benchmark and applying the QNLP pipeline to it would be a valuable

interdisciplinary contribution, particularly within the Asian regional data-science context that motivates this work.

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Declaration of Conflicting Interests

The authors declare that there is no conflict of interest regarding the publication of this paper.

Data Availability

The Stanford Sentiment Treebank is publicly available from Socher et al. (2013). The code, configurations, training logs and per-seed prediction artefacts produced in this work are released under the MIT licence at https://github.com/hublinhhdn/nlp_quantum_hub.

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